Name: Shriniketh Kana
SRN: PES1UG24CS446
Section: H
Course: Software Engineering Lab 3
Problem Statement #18: Clinical Trial Subject Recruitment Engine

Architecture Selection and Justification

Architecture Selection: We chose Layered Architecture for the Clinical Trial Subject Recruitment Engine System.

Styles considered: Microservices was rejected because candidate, consent and eligibility records must stay consistent with one another, and splitting them across independently deployed services brings distributed transactions that a regulated clinical system cannot tolerate. Plain Client-Server was rejected because it leaves the eligibility rules, consent handling and data access mixed in one tier.

Reason 1: The recruitment workflow is a staged sequence (register, submit pre-screening survey, check eligibility, give consent, schedule visit) and every step follows the same top-down path: a portal calls the Recruitment Manager, which calls the Eligibility Rule Engine or the Consent and Audit Service, which read and write through the Data Access Component. The scenario maps directly onto the layers.

Reason 2: Trial protocols change often. When inclusion or exclusion criteria are amended, only the Eligibility Rule Engine changes, because it is reached solely through the IEligibilityCheck interface, so the portals and the Recruitment Manager are untouched. Likewise the Data Access Component hides the database schema behind IDataQuery and IProtocolData.

Security Advantage: Survey responses and consent records are protected health information, so the presentation layer is never allowed to query the database directly. Every request from the Candidate Portal and the Researcher Console enters through the Recruitment Manager, giving one place to enforce authentication, role-based authorisation and input validation. The Consent and Audit Service sits on that same path, so consent checks and audit entries cannot be bypassed, and only the data layer holds database credentials.

Performance Benefit: Eligibility checking is the heaviest operation, since each check evaluates a full survey against all criteria of a protocol. The Data Access Component caches protocol criteria and reuses pooled connections for every caller above it, so a researcher screening a batch of candidates loads the criteria once rather than once per candidate. Calls between layers are in-process, adding no serialisation or network latency.